

Using AI to Improve Supply Chains in Oregon

Final Handoff Report

Caden Smith, David Foster, Luke Garci
CS 463 Spring 2026
Oregon State University

Contents

1. Introduction
2. Current Project Status
3. Technologies, Libraries, APIs, and Tools Used
4. Deviations from the Development plan
5. Major Challenges and Solutions
6. Project Continuance
7. Conclusion

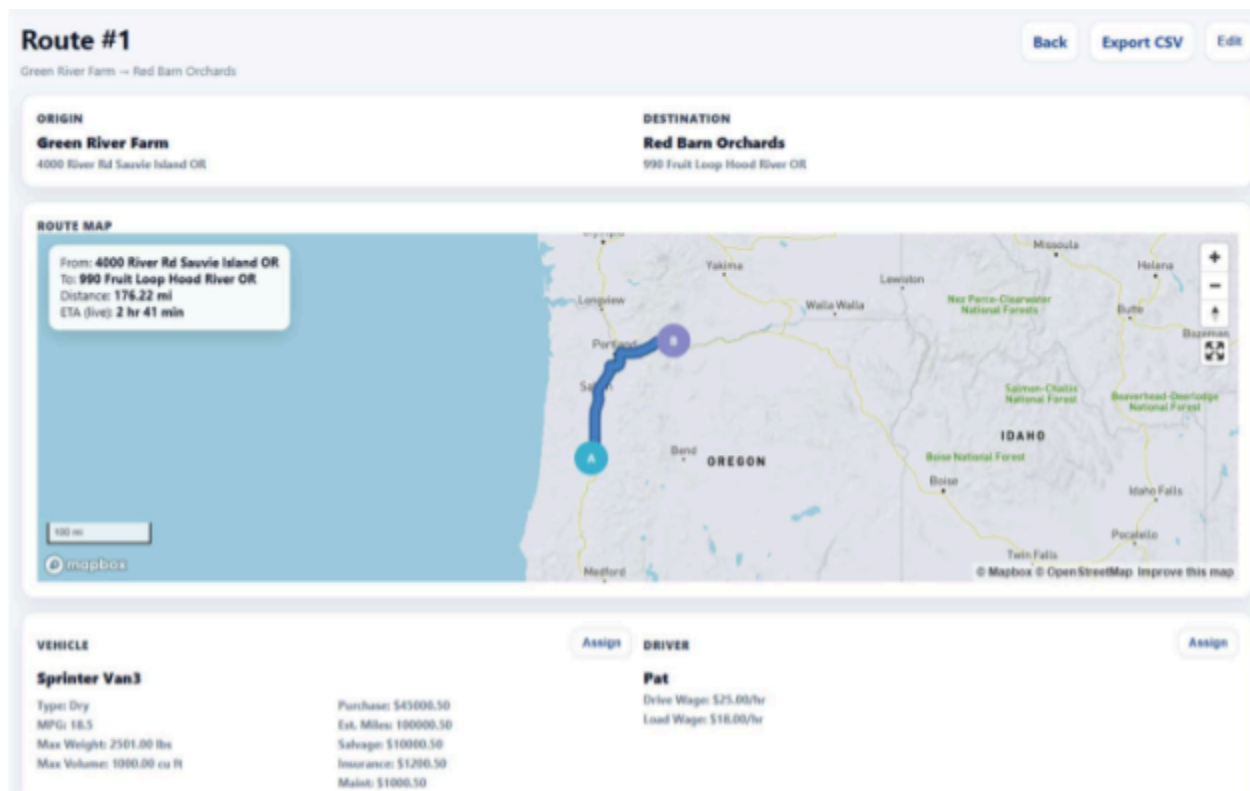
Introduction

This report is for the final handoff for our project Using AI to Improve Oregon Supply Chains, which was a capstone project created while working with our project partner Jim Cupples who works at the Oregon Department of Agriculture. The main goal of the project was to build a useful tool for agricultural stakeholders specifically small to medium sized farms in Oregon who need a better way to understand/visualize transportation costs, route profitability, and delivery planning across the local food supply chain.

The project became a web application focused on helping users model delivery routes between farms, food hubs, stores, and other locations. Users can enter locations, vehicles, drivers, products, route information, and manifest items. From there, the application calculates transportation costs, total route costs, product-level profitability, and break-even information. The project is intended to help users make more informed decisions instead of estimating transportation costs manually or relying on spreadsheets that may not fully account for fuel, driver wages, depreciation, maintenance, insurance, and product margins. This report is written for the next team or developer who may continue the project. The goal is to explain what currently works, what still needs more attention, what technologies were used, to be in a better place to quickly continue developing after the handoff.

Current Project Status

The project is currently in a functional prototype stage and has a working route dashboard, route details pages, account creation, login, anonymous mode, asset management, manifest management, and cost calculation features. The main dashboard allows users to view existing routes, see route names, origin and destination information, products in the manifest, revenue, total cost, and net profit information.



Dashboard showing the route visualized in Map API, and associated costs

Users can create an account, set up TOTP authentication, create routes, manage locations,

manage vehicles, manage drivers, manage products, and manage route loads. Users can also choose to use this application anonymously, with an option to make an account later on. This was an important feature because one project goal was to reduce barriers for farmers or other users who might not want to create an account before trying the tool. Anonymous users can use the calculator temporarily, and then upgrade their account later without losing entered data. The anonymous accounts are valid for 7 inactive days with the timer resetting once the user uses the application again and can be upgraded into permanent accounts while preserving data.

The route details page is one of the most important parts of the application. It shows route overview information, origin and destination addresses, a map, truck and driver details, financial metrics, time and rate information, a cost breakdown, load stats, profitability, trip break-even information, and manifest-level calculations.

FINANCIAL METRICS

Recalculate Costs

Time & Rates

Drive Mins

Load Mins

Unload Mins

Distance

Gas Price

345.73

60.0

30.0

352.4 mi

\$3.50/gal

Cost Breakdown

Driver (Drive)

Driver (Load)

Driver (Unload)

Total Driver

\$144.05

\$18.00

\$9.00

\$171.05

Fuel

Depreciation

Daily Insurance

Daily Maintenance

\$66.68

\$15.20

\$4.17

\$3.48

COGS

\$15.62

Load Stats

Items

Total Weight

Total Volume

1

100.0 lbs

6.25 cu ft

Profitability (Entered vs Calculated)

Entered Revenue

Calculated Revenue

\$1500.00

\$374.38

Total Cost: \$276.20

Total Cost: \$276.20

Profit: \$1223.80

Profit: \$98.18

Margin: 81.6%

Margin: 26.2%

Trip Break-Even

+\$98.18

Net Trip Profit

Delivery consumes 72.6% of total margin (\$260.58 of \$358.76)

Financial Metrics visualized associated with a given route giving insight to the costs, and potential revenue/loss for a route.

These features connect closely to the original goal of helping users understand the real cost of transporting food products. Several major requirements are functional. Users can create and

manage locations, vehicles, drivers, products, and loads. The application calculates fuel cost, driver cost, depreciation, insurance, maintenance, total cost, revenue, profit, and margin. It also supports allocating delivery cost by weight or volume across manifest items and showing break-even information per product. We focused heavily on profit margins, break-even calculators, transport costs, dashboard alerts, QA, code cleanup, and preparing the code for handoff towards the end of the project with some areas still not fully production-ready. The application works as a prototype, but future teams should treat the project as something that needs more testing, cleanup, deployment work, and user feedback before being used as a long-term production system. The test plan specifically calls out areas of concern such as financial calculation accuracy, Mapbox failure handling, multi-tenant data isolation, authentication/session security, form validation, and test coverage.

Technologies, Libraries, APIs, and Tools Used

The application was built as a Flask/MySQL web application. Flask was used for the backend web framework, while MySQL handled persistent storage. The project uses a multi-tenant database structure so that different users or organizations can keep their data separate. Every table uses a composite tenant-based structure, and database access functions are expected to filter by tenant ID from the JWT. This is one of the most important technical areas for future

teams to understand because any tenant isolation bug could expose one user's data to another user.

The frontend is made with standard web technologies, including HTML, CSS, and Flask templates. A lot of the frontend work focused on making the application usable rather than overly complex. We built forms for route creation, product entry, vehicle entry, driver entry, location entry, route detail views, anonymous mode banners, and CSV import screens. We also included product entry fields, driver information templates, location templates, vehicle entry templates, and route detail pages. The project uses the Mapbox API to calculate trip distance and drive time between route locations. This was a strong solution because distance and time are core parts of transportation cost estimation. However, Mapbox is also one of the more sensitive dependencies. Such as how the application calls Mapbox during cost calculation, and if the API fails or returns no result, the app may default to 0 miles and 0 minutes. That means distance-based costs can become zero without clearly warning the user. Future teams should improve this by showing a visible warning when route distance cannot be calculated.

Authentication uses JWT cookies and TOTP two-factor authentication. New users must set up an authenticator app after registration. The quick start guide explains setup through Google Authenticator, Microsoft Authenticator, or manual setup using a 32-character key. TOTP is also used for password recovery so users can reset their password without being locked out. The project also used Trello for task management, GitHub for source control, and progress reports/sprint planning documents for tracking team work. The Trello board was used to organize sprint tasks such as anonymous mode, password recovery, CSV imports, break-even calculators, QA, and deployment preparation.

Deviations from the Development Plan

The project did change from the original development direction. At the start, we were still figuring out how broad the tool should be and how much AI-related work would be practical within the capstone timeline. As the project moved forward, we focused more on building a strong transportation cost calculator and supply-chain decision support tool instead of trying to force AI into every feature. This was a good change as the project partner needed something useful and understandable and that could be put in front of real users to gather feedback. A working route calculator with clear financial outputs is more valuable than an unfinished AI feature that users cannot trust. The final product still supports the broader idea of improving supply-chain decisions, but it does so through structured data, cost modeling, route calculations, and reporting.

Another deviation was the addition of anonymous mode. This became important because farmers and small agricultural businesses may not want to create an account right away. Adding anonymous mode made the tool easier to test and reduced friction for first-time users. This was implemented due to valuable feedback and anonymous mode, account upgrades, and temporary tenant handling became priority in development. CSV import and export also became more important over time since agricultural users may already have product, route, or asset information in spreadsheet form. Future teams should continue improving CSV workflows because this could make the tool much easier to adopt and distribute.

Major Challenges and Solutions

One major challenge was connecting frontend forms to backend logic in a way that stayed organized. The project has many related objects: routes, locations, vehicles, drivers, products, and manifest items. Each of these needed forms, validation, storage, and display logic. We solved this by creating dedicated management overlays and pages for different asset types. Another challenge was financial calculation accuracy. Small mistakes in depreciation, insurance, fuel, or driver wage logic could make the tool misleading. The test plan identifies this as a major risk and recommends white-box unit testing for calculation functions such as depreciation, insurance, driver costs, fuel costs, safe parsing helpers, and manifest aggregation.

Authentication was also a challenge, and adding account registration, login, password reset, TOTP setup, JWT cookies, and anonymous mode. These features improve usability and security, but they also create more edge cases. Future teams should continue testing expired tokens, tampered tokens, anonymous sessions, password recovery, and users who have not completed TOTP setup. The final major challenge was documentation. We made progress reports, sprint planning documents, a quick start guide, and a test plan, but future teams should still improve the project wiki and technical setup documentation. Progress report notes mention that documentation and wiki updates were an area that needed more attention.

Project Continuance

The most important next step is to prepare the project for a future team to continue development. The current project is useful, but it should not be treated as finished production software as it needs more testing, stronger error handling, cleaner documentation, and deployment work. Future work should start with the financial calculation engine with every formula should be covered by automated tests, especially depreciation, insurance, maintenance, fuel, driver wages, entered revenue, calculated revenue, total cost, profit, and margin. These calculations are the core value of the application, so they need to be correct and easy to verify. The next team should also improve Mapbox error handling as right now if Mapbox fails, the app may use 0 miles and 0 minutes. That can make a route look cheap and be misleading to a user we expect to not be completely technically literate. Instead the app should display an error or warning and prevent the user from trusting incomplete cost results.

Multi-tenant security is another high priority and future teams should test tenant isolation carefully. A user should never be able to view, edit, or delete another tenant's routes, locations, vehicles, drivers, products, or manifest items. The test plan already identifies IDOR testing, altered JWT testing, and tenant-scoped database tests as important. Deployment is also still an important handoff item. The team discussed packaging and migrating the project to GCP or the sponsor's preferred environment. This should be completed carefully, including environment variables, database setup, Mapbox keys, secret keys, HTTPS, backups, and production logging. Finally, future teams should keep the user audience in mind. The tool is meant for agricultural stakeholders, not just developers. That means the interface should stay simple, instructions should be clear, and error messages should be friendly.

Conclusion

Overall, this capstone project taught us a lot about building software for a real partner with real user needs. One of the biggest lessons was that useful software does not always need to be overly complicated and the most valuable part of this project was not just adding advanced features, but making transportation costs more understandable for people who may not have time to manually calculate every detail. The team also learned how important communication is by conducting Sprint planning, Trello tasks, progress reports, and partner feedback and seeing how valuable it was to keep the project moving, and iterating over important feedback. When the team stayed organized, development went much smoother. When documentation fell behind, it became harder to remember every decision and explain the system clearly.

The project gave us experience with full-stack development, authentication, database design, API integration, cost modeling, testing plans, and handoff documentation. It also showed us how quickly complexity grows when a project includes users, accounts, anonymous sessions, route data, vehicles, drivers, products, CSV workflows, and financial calculations. The biggest takeaway is that future development should focus on reliability, testing, and usability. The application already has a strong foundation, but the next team should harden the calculation engine, improve deployment, expand documentation, and continue testing with real agricultural stakeholders. If those pieces are improved, this project could become a useful tool for helping

Oregon food supply-chain participants understand delivery costs and make better decisions and allow them to remain competitive in this everchanging environment.